



# **BUSINESS LOGIC SPECIFICATION**

Automatic Production Line Analytics

**Version 1.0**

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## BUSINESS OBJECTIVES

Automatic Production Line Analytics

### BUSINESS OBJECTIVES



- Monitor production efficiency
- Measure OEE
- Compare plan vs actual
- Monitor quality
- Support operational decisions
- Enable continuous improvement

### BUSINESS QUESTIONS



- Current OEE?
- Low performing lines?
- Production losses?
- Plan vs Actual?
- Which KPIs need attention?



# MANUFACTURING KPI FRAMEWORK

Automatic Production Line Analytics

## KPI FRAMEWORK



### INDUSTRIAL LOAD



### VOLUME LOAD



### PERFORMANCE



### QUALITY



### AVAILABILITY



### OEE



## KPI BUSINESS RULES



$$\frac{\text{Available positions}}{\text{Total positions}}$$

$$\frac{\text{Planned Positions}}{\text{Available positions}}$$

$$\frac{\text{Actual Production}}{\text{Planned Production}}$$

$$\frac{\text{Good Production}}{\text{Total Production}}$$

$$\frac{\text{Actual Production Time}}{\text{Planned Production Time}}$$

$$\text{Industrial Load} \times \text{Volume Load} \times \text{Availability} \times \text{Performance} \times \text{Quality}$$

## BUSINESS MEANING



Technically approved  
all existing  
production positions

Available equipment  
is loaded by client  
demands

Production at  
expected speed

Scrap generation

Effective use of planned  
production time

Efficient and stable  
production process



# BUSINESS ASSUMPTIONS & LIMITATIONS

Automatic Production Line Analytics

## ASSUMPTIONS



- Planning data are accurate
- Scrap data are complete
- Production parameters are maintained

## LIMITATIONS



- No downtime tracking
- No maintenance records
- No machine alarms
- Root cause analysis requires additional data



## KEY TAKEAWAYS

Automatic Production Line Analytics

### SUMMARY

- Standardized KPI definitions
- Transparent business rules
- Reusable calculation logic
- Consistent reporting
- Scalable analytical model
- Supports continuous improvement